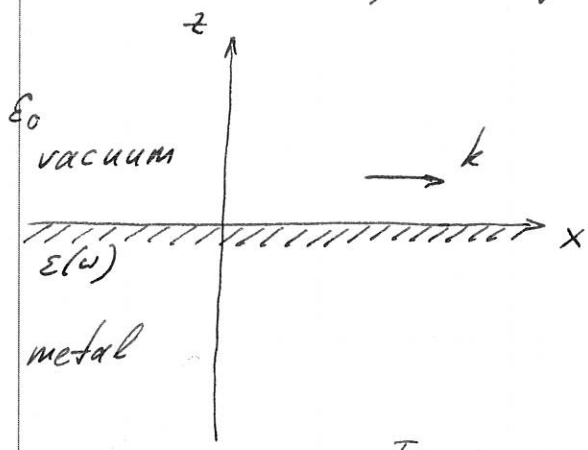
[redacted] plasma waves on the surface of a metal.



Maxwell equations in the absence of free charges

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

In a non-magnetic medium $\vec{B} = \mu_0 \vec{H}$ (or $\vec{H} = \vec{B}/\mu_0$)

For a harmonic wave,

$$\vec{E}, \vec{D}, \vec{B} \sim e^{-i\omega t} \quad \text{and we get}$$

$$\nabla \times \vec{E} = i\omega \vec{B} \qquad (1)$$

$$\nabla \times \vec{B} = -i\mu_0 \omega \vec{D} = -i\mu_0 \omega\, \varepsilon(\omega) \vec{E} = -\frac{i\omega}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0} \vec{E} \qquad (2)$$

Substituting $\vec{B}$ from the first equation to the second one and using $(\nabla \cdot \vec{E} = 0)$ $\nabla \times (\nabla \times \vec{E}) = -\nabla^2 \vec{E}$ we find the equation for $\vec{E}$ only:

$$\nabla^2 \vec{E} + \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0} \vec{E} = 0 \qquad (3)$$

Following our understanding that bulk plasma waves are longitudinal, we expect a surface plasma wave to have a longitudinal (E_x) component:

$$E_x(x,z) = f(z) e^{ikx}$$

$f(z)$ satisfies (see Eq. (3)) the equation

$$f'' - \left(k^2 - \frac{\omega^2}{c^2} \frac{\varepsilon(\omega)}{\varepsilon_0}\right) f = 0$$

Note that $k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}$ must be ~~negative~~ positive for $f(z)$ to have exponentially decaying solutions

$$f(z) \sim \exp\left(z\sqrt{k^2 - \frac{\omega^2}{c^2}\frac{\varepsilon(\omega)}{\varepsilon_0}}\right) \quad , \quad z<0$$

Similarly, we choose exponentially decaying solution for $z>0$ (where $\varepsilon(\omega) \equiv \varepsilon_0$).

Using the condition of continuity of E_x we finally find

$$E_x = Ae^{ikx} \begin{cases} \exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right), & z>0 \\ \exp\left(z\sqrt{k^2 - \frac{\omega^2\varepsilon}{c^2\varepsilon_0}}\right), & z<0 \end{cases}$$

We now find E_z from the condition that

$$\nabla\cdot\vec{E}=0 \implies \frac{\partial E_x}{\partial x} + \frac{\partial E_z}{\partial z} = 0 \implies ikE_x + \frac{\partial E_z}{\partial z} = 0$$

or $\quad E_z = -ik\int dz\, E_x$

$$E_z(z>0) = \frac{ikAe^{ikx}}{\sqrt{k^2 - \frac{\omega^2}{c^2}}}\exp\left(-z\sqrt{k^2 - \frac{\omega^2}{c^2}}\right),$$

and

$$E_z(z<0) = -\frac{ikAe^{ikx}}{\sqrt{k^2 - \frac{\omega^2\varepsilon}{c^2\varepsilon_0}}}\exp\left(z\sqrt{k^2 - \frac{\omega^2\varepsilon}{c^2\varepsilon_0}}\right)$$

Note that the integration constant is chosen in such a way that $E_z \to 0$ when $z \to \pm\infty$.

Finally, from the continuity of D_z we obtain the second boundary condition

$$\varepsilon_0 E_z(z=0+) = \varepsilon E_z(z=0-):$$

$$\frac{\varepsilon_0}{\sqrt{k^2-\frac{\omega^2}{c^2}}} = \frac{-\varepsilon}{\sqrt{k^2-\frac{\omega^2\varepsilon}{c^2\varepsilon_0}}} \Rightarrow \text{need } \boxed{\varepsilon(\omega)<0}$$

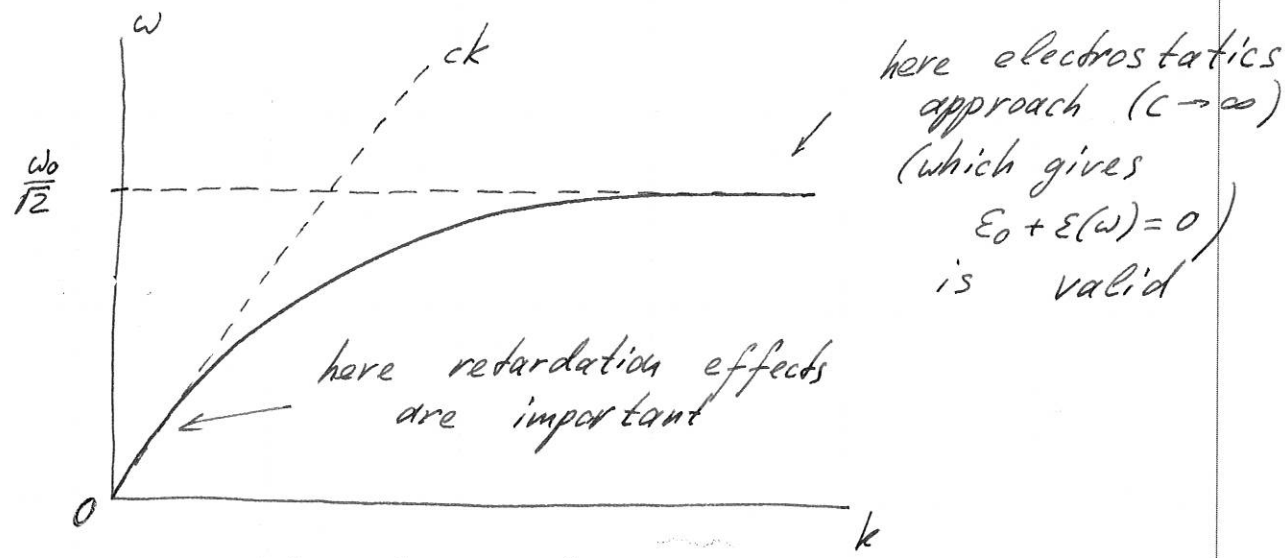
or,
$$k^2-\frac{\omega^2}{c^2}\frac{\varepsilon}{\varepsilon_0} = \left(k^2-\frac{\omega^2}{c^2}\right)\frac{\varepsilon^2}{\varepsilon_0^2} \qquad k^2\left(1-\frac{\varepsilon^2}{\varepsilon_0^2}\right) = \frac{\omega^2}{c^2}\left(\frac{\varepsilon}{\varepsilon_0}-\frac{\varepsilon^2}{\varepsilon_0^2}\right)$$

$$k^2 = -\frac{\omega^2}{c^2}\,\frac{\varepsilon^2/\varepsilon_0^2-\varepsilon/\varepsilon_0}{1-\varepsilon^2/\varepsilon_0} = +\frac{\omega^2}{c^2}\,\frac{\varepsilon/\varepsilon_0}{1+\varepsilon/\varepsilon_0}$$

Using now the Drude formula $\varepsilon/\varepsilon_0 = 1-\frac{\omega_0^2}{\omega^2}$

we find
$$\boxed{k^2 = \frac{\omega^2}{c^2}\,\frac{\omega_0^2-\omega^2}{\omega_0^2-2\omega^2}} \qquad (4)$$

Sketch



Note that the second solution of bi-quadratic equation (4) has $\omega > \omega_0$ and does not correspond to a surface wave.

$$(\omega_p^2 - 2\omega^2)c^2k^2 = \omega_p^2\omega^2 - \omega^4$$

$$\omega^4 - \omega^2(\omega_p^2 + 2c^2k^2) + \omega_p^2c^2k^2 = 0$$

$$\omega^2 = \frac{1}{2}\left(\omega_p^2 + 2c^2k^2 \pm \sqrt{(\omega_p^2+2c^2k^2)^2 - 4\omega_p^2c^2k^2}\right)$$

$$= \frac{1}{2}\left(\omega_p^2 + 2c^2k^2 \pm \sqrt{\omega_p^4 + 4c^4k^4}\right)$$

1) $k \to 0$ limit: $\omega_p^2\sqrt{1 + \frac{4c^4k^4}{\omega_p^4}} = \omega_p^2 + \frac{2c^4k^4}{\omega_p^2}$

$$\omega^2 = \frac{1}{2}\left(\omega_p^2 + 2c^2k^2 \pm \left[\omega_p^2 + \cancel{\frac{2c^4k^4}{\omega_p^2}}\right]\right)$$

$\ominus$ $= c^2k^2$ [for $\oplus = \omega_p^2 + c^2k^2$ ← describes wave propagating in the conductor]

2) $k \to \infty$ limit: $2c^2k^2\sqrt{1 + \frac{\omega_p^4}{4c^4k^4}} = 2c^2k^2 + \frac{\omega_p^4}{4c^2k^2}$

$$\omega^2 = \frac{1}{2}\left(\omega_p^2 + 2c^2k^2 \pm \left[2c^2k^2 + \frac{\omega_p^4}{4c^2k^2}\right]\right)$$

$\oplus$ $= \frac{1}{2}\omega_p^2 + 2c^2k^2$

$\ominus$ $= \frac{1}{2}\omega_p^2 - \frac{\omega_p^4}{8c^2k^2} \approx \left(\frac{\omega_p}{\sqrt{2}}\right)^2$

Surface wave corresponds to $\ominus$ sign above.

Also, the same can be seen by checking the requirement

$\varepsilon(\omega) < 0 \Rightarrow 1 - \frac{\omega_p^2}{\omega^2} < 0 \Rightarrow \omega < \omega_p.$

Clearly only $\ominus$ root satisfies this condition since

$\omega_-(k) \leq \frac{\omega_p}{\sqrt{2}}$ for all k.